

Work Experience

Machine Learning Engineer 3D Scan-to-BIM Project	mVizn Pte. Ltd. Singapore (Remote)	Jan 2024 – Present
- Led Backend Engineering efforts, including API Design and System Architecture, and deployed production AI systems using Python, Docker, and GCP (VMs, Cloud Storage, MySQL). - Achieved a 60% improvement in client system performance through scalable system design, demonstrating the ability to ship and maintain reliable production systems. - Implemented Monitoring and Evaluation frameworks by deploying a dashboard to track and visualize ML model outputs. - Resolved over 10 critical post-deployment issues by troubleshooting AI model behavior and collaborating with stakeholders to align technical solutions with business requirements. - Developed a comprehensive 3D Computer Vision pipeline for semantic segmentation of 3D point clouds using Python and PyTorch.		
Machine Learning Engineer Advanced driver-assistance system (ADAS)	Mercedes-Benz Research and Development India Bengaluru, India	Aug 2018 – Aug 2019
- Developed a complete 2D computer vision pipeline for Vulnerable Road User (VRU) detection: Trained Yolo v3 model with in-house annotated images, improving Level 3 ADAS vehicle capabilities by 30%. - Built and automated evaluation frameworks and quality checks using Python scripts, increasing annotation throughput by 20% and improving data accuracy by 25%.		
Systems Engineer Ultimatix Team	Tata Consultancy Services Kolkata, India	Aug 2014 – Aug 2015
- Maintained and enhanced high-availability backend systems (Java, MySQL) for TCS's internal website, ensuring seamless functionality for over 100,000 users (demonstrating experience with scale and reliability). - Collaborated with cross-functional teams to streamline the feature release cycle, reducing deployment time by 20%		

Skills

Languages:	Python, C++, SQL
AI/Agent Systems:	Multi-Agent Systems, Agent Orchestration (Crew AI, Google ADK), Tool Calling/Execution, Shared State/Memory, Production AI Systems, LangChain, LLM APIs (Gemini, Claude)
Cloud & Ops:	Google Cloud Platform (GCP) (Cloud Run, Vertex AI), Git, Docker
Other:	Scalable System Design, Backend Engineering, System Architecture, Data Structures & Algorithms, Workflow Orchestration

Education

PhD, Jadavpur University, Kolkata, India.	2019–2024
M.Tech. Computer Science & Engineering, Jadavpur University, Kolkata, India. GPA – 8.83	2016–2018
B.Tech. Computer Science & Engineering, Techno India, Kolkata, India. GPA – 8.81	2010–2014

Projects

- Doctor Appointment Scheduler** (Mar '26 – Apr '26) – Architected and deployed a scalable, production-level Multi-Agent System for appointment scheduling. Designed the Agent Orchestration and coordination, integrating intelligent doctor matching and calendar scheduling (task hand-off/execution). Implemented robust Tool Calling and Execution using Composio for calendar actions and Google Maps MCP. The scalable Backend Engineering architecture utilizes a RESTful API (FastAPI) served on Google Cloud Run for containerized deployment, integrating Google Cloud SQL and Vertex AI.
GitHub – https://github.com/arjitde92/Doctor_Appointment_Scheduler
Live Link – <https://doctor-appointment-app-173427564927.asia-south1.run.app/>
- Job Application Agent** (Apr '25 – May '25) – Developed a real-world Multi-Agent System for automation, utilizing Crew AI for advanced workflow orchestration of specialized agents. Engineered Shared

State/Memory by integrating Google BigQuery (vector store) and Vertex AI (embeddings) for consistent, scalable semantic search and storage across agents. Implemented custom tools (LinkedIn/GitHub scrapers) and leveraged LangChain for core system logic, all built in Python.

GitHub - https://github.com/arijitde92/Job_Application_Agent

- **Online Assignment Submission Portal** (Oct '24 - Nov '24) - A Flask + MySQL WebApp for submission & evaluation of C programming assignments built in a docker container. Teachers can create assignments whereas students can view assignments, submit code & get it evaluated with an in-built GCC compiler. Live link - <https://subsequent-heath-arijit-home-3b9d0400.koyeb.app/>
Github - https://github.com/arijitde92/Online_Programming_Assignment_Portal
- **Spiritual Chatbot** (Jan'24 - Feb'24) - An LLM chatbot that personifies a Hindu God. Uses OpenAI GPT-3.5 turbo model APIs and Langchain for creating prompt templates. Attempted prompt engineering techniques to make chatbot stay within context and purpose. The front end was built using streamlit while python was at backend. Live link - <https://spiritualchatbot.streamlit.app/> Github - <https://github.com/arijitde92/SpiritualChatBot>
- **3D Brain Analysis and Visualization App** (Jul'24 - Aug'24) - A Flask Application that can segment the hippocampus or a brain tumor from a 3D brain MRI scan. The segmentation models are built using Hippocampus Segmentation Factory (HSF) for Hippocampus Segmentation and a 3D Attention UNet model for brain tumor segmentation. Github - https://github.com/arijitde92/Brain_Seg_App

Publications

1. **A. De**, A. Santra, M. Tiwari and A. S. Chowdhury, "Predicting Genetic Markers for Brain Tumors Using a Composite Loss," in *IEEE Transactions on Computational Biology and Bioinformatics (TCBB)*, 2025 July. doi: 10.1109/TCBBIO.2025.3593318.
2. **A. De**, A. S. Chowdhury, (2025). Shape Induced Multi-class Deep Graph Cut for Hippocampus Subfield Segmentation. In: Antonacopoulos, A., Chaudhuri, S., Chellappa, R., Liu, CL., Bhattacharya, S., Pal, U. (eds) Pattern Recognition. (**ICPR 2024**). Lecture Notes in Computer Science, vol 15313. Springer, Cham. https://doi.org/10.1007/978-3-031-78201-5_16
3. **A. De**, N. Das, PK. Saha, A Comellas, E Hoffman, S Basu, T Chakraborti. MSO-GP: 3-D segmentation of large and complex conjoined tree structures. **Methods. 2024** June 3;229:9-16. doi: 10.1016/j.ymeth.2024.05.016. Epub ahead of print. PMID: 38838947.
4. **A. De**, P. Ebenezer, "Sleep Apnea sub-type detection from Polysomnography signals," 2024 IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (**IATMSI 2024**), Gwalior, India, 2024, pp. 1-6
5. **A. De**, M. Tiwari, & A. S. Chowdhury. 3D Hippocampus Segmentation Using a Hog Based Loss Function with Majority Pooling. In 2023 IEEE International Conference on Image Processing (**ICIP**), Kuala Lumpur, Malaysia **2023**, October, pp. 2260-2264.
6. **A. De**, R. Mhatre, M. Tiwari and A. S. Chowdhury, "Brain Tumor Classification from Radiology and Histopathology using Deep Features and Graph Convolutional Network," 2022 26th International Conference on Pattern Recognition (**ICPR 2022**), Montreal, QC, Canada, 2022, pp. 4420-4426
7. **A. De**, M. Tiwari, E. Grisan, A.S. Chowdhury, "A Deep Graph Cut Model for 3D Brain Tumor Segmentation", 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (**EMBC 2022**).
8. **A. De**, A.S. Chowdhury, "DTI based Alzheimer's Disease Classification with Rank Modulated Fusion of CNNs and Random Forest", **Expert Syst. Appl.** 169 (2021), 114338. DOI: 10.1016/j.eswa.2020.114338

Open Source Contributions

- Contributed to [AI-Code](#) as part of Social Summer of Code (**SSoC**) 2024 & [DL-Simplified](#) as part of Social Winter of Code (**SWoC**) 2023 Hackathons.
- Contributed to [nexB](#) and [TheAlgorithms](#) repositories pertaining to both feature enhancements and documentations.

Certifications

- Deep Learning, a 5-course specialization, by Deeplearning.ai. Verify [here](#).
- MCPS: Microsoft Certified Professional, Microsoft Certification No.: 1042C4-5DHF30